

PED (15) 3022 -FLUID MECHANICS AND PNEUMATICS

MODEL QUESTION PAPER

MAX MARKS :100

TIME :3 Hrs

Part A

(answer all questions in one or two sentences, carries 2 marks each.)

I

1. Define Hydraulics
2. State Pascals Law
3. Differentiate between Laminar & Turbulant flow
4. List various types of Pumps
5. Sketch the symbol of Pneumatic 4/3 DC Valve.

Part B

(answer any five questions , carries 6 marks each. (5x6=30marks))

II

1. Define sp. Weight, mass density, viscosity
2. Sketch and briefly explain Bourdon's tube pressure gauge.
3. Describe about the energy losses in Pipes.
4. Derive the Continuity Equation.
5. Briefly explain the working of Hydraulic Gear Pump.
6. Compare Hydraulic and Pneumatic Power.
7. Briefly explain the working of Pneumatic cylinder.

Part C

(answer four full questions from each Module, (15x4= 60 marks))

Module I

- III
- a) Calculate sp. Weight, mass density, Weight of one Litre of Petrol having sp. Gravity 0.7. (6)
 - b) A differential Manometer is used to measure the difference of pressure of oil of sp. Gravity 0.8, contained in two pipes at same level. If the deflection of the mercury Manometer is 100mm. Determine the difference of pressure of oil in the two pipes in Kpa. (9)

OR

- IV
- a) A rectangular pontoon is 5m long, 3m wide and 1.2m height. The depth of immersion of the pontoon is 0.8m in Sea water. If the centre of gravity is 0.6m above the bottom of the pontoon, Determine the metacentric height. Assume the density of Sea water is 1025 Kg/m^3 (6)
 - b) A circular plate 4m in Dia is immersed in Sea water having sp. Gravity 1.01 such that its circular surface makes an angle of 30 degrees with the free surface and its top edge is 1.2m below it. Find the total pressure. (9)

Module II

- V a) Derive Bernoulli's Theorem (6)
 b) A horizontal venturimeter 150x75mm is used to measure flow rate of water. Determine the deflection of mercury in the U tube manometer, If the flow rate is 35 lit/s. Take $C_d=0.96$ (9)

OR

- VI a) Explain the purpose of Notches and its classification (6)
 b) Determine the Dia of pipe line 2 Km long, If the loss of head is 10m of water. The velocity with which the water flows through the pipe line is 0.75 m/s. Take $f=0.01$ (9)

Module III

- VII a) Describe about Fluid power and its applications (6)
 b) With a neat sketch describe the working of intensifier and weight loaded accumulator (9)

OR

- VIII a) Explain with sketch the Hydraulic linear actuator. (6)
 b) Draw the circuit diagram of Hydraulic robotic arm and explain (9)

Module IV

- IX a) Describe about Pneumatic Muffler and its applications (6)
 b) Draw the acceleration and retraction circuit diagram in Pneumatics (9)

OR

- X a) Explain with neat sketch the Solenoid valve. (6)
 b) Draw the circuit diagram of sequencing of two cylinders by electro pneumatic system (9)
